

Rational Hybrid Monte Carlo of **SIMULATeQCD**

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Here we describe the parameters of the the rational hybrid Monte Carlo (RHMC) [1, 2] algorithm used to generate HISQ quarks with **SIMULATeQCD**. They are given in Table 1. For more information about these parameters and how the RHMC is implemented, we refer to Refs. [3, 4].

References

- [1] K. Clark and A. D. Kennedy. The RHMC algorithm for 2 flavours of dynamical staggered fermions. *Nuclear Physics B - Proceedings Supplements*, 129-130:850–852, 2004.
- [2] K. Clark. The Rational Hybrid Monte Carlo algorithm. In *Proceedings of XXIVth International Symposium on Lattice Field Theory — PoS(LAT2006)*, page 004, Tucson, Arizona, 2006. Sissa Medialab.
- [3] L. Mazur et al. **SIMULATeQCD**: A simple multi-GPU lattice code for QCD calculations. *Computer Physics Communications*, 300:109164, 2024.
- [4] L. Mazur et al. **SIMULATeQCD** documentation. <https://latticeqcd.github.io/SIMULATeQCD/>, 2025. Accessed: 2026-04-28.

XML Tag	Explanation
<code>always_acc</code>	Metropolis (0); always accept proposal (1)
<code>cgMax</code>	maximum CG steps
<code>load_conf</code>	hot (0); cold (1); load (2)
<code>no_md</code>	number of molecular dynamics steps in trajectory
<code>no_pf</code>	number of pseudo-fermion fields
<code>no_step_sf</code>	number of steps for strange-quark integration
<code>no_sw</code>	number of steps for gauge integration
<code>rand_flag</code>	new random numbers (0); read in (1)
<code>seed</code>	seed for RNG
<code>step_size</code>	step size of trajectory

Table 1: Input parameters to the RHMC and their meaning.